Project 1: Retail - Customer Behavior & RFM Analysis

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Role : Data Analyst

➢ Project Title

Consumer360 – Retail Analytics & Customer Churn Intelligence System

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1. Project Overview

Consumer360 is an end-to-end retail analytics project developed during my internship at **Infotact Solutions**. The project converts raw retail transaction data (100,000+ rows) into actionable business insights using SQL, Python, and Power BI.

The system performs customer behavior analysis using **RFM segmentation**, identifies **churn-risk customers**, and provides interactive dashboards for sales and customer intelligence. The pipeline is automated using Python and Task Scheduler to simulate a real-world analytics workflow.

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2. Objectives

- Analyze retail sales and customer data
- Segment customers using RFM model
- Identify churn-risk customers
- Build interactive BI dashboards
- Implement Row Level Security (RLS)
- Automate RFM pipeline using Python

❖ Tools & Technologies

- Python (Pandas)
- SQL (PostgreSQL)
- Power BI
- DAX
- Star Schema Modeling
- RFM Analysis
- Task Scheduler Automation
- Row Level Security

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3. Dataset Description

- Source: Retail transaction dataset
- Size: ~100,000 rows
- Fields: Orders, Customers, Products, Amount, Region, Dates
- Data Type: CSV → PostgreSQL → Power BI

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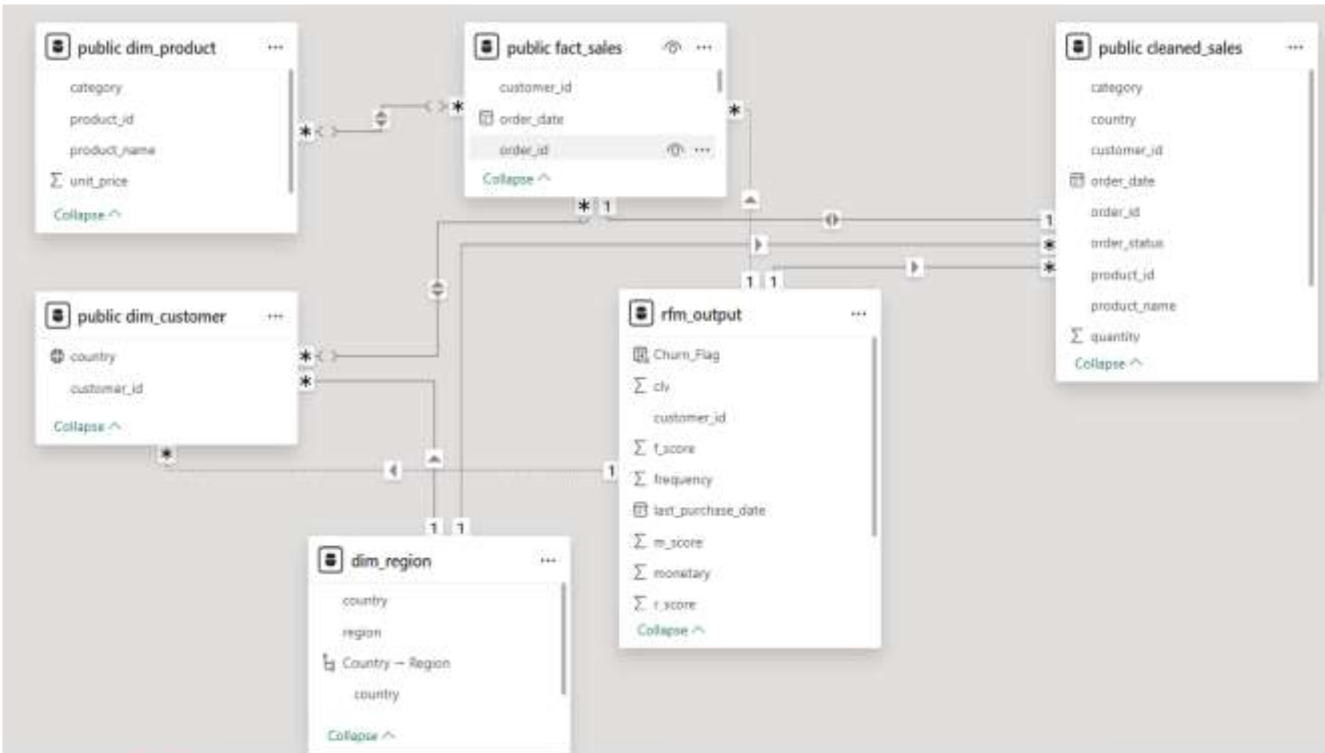
4. Data Modeling (SQL Layer)

A **Star Schema** model was created in PostgreSQL:

- Fact Table: Sales
- Dimension Tables: Customer, Product, Region
- Relationships created for analytics optimization

This structure improves query performance and dashboard clarity.



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5. Data Processing & RFM Analysis (Python)

Python was used to:

- Clean and standardize columns
- Convert date fields
- Calculate Recency, Frequency, Monetary
- Generate RFM scores
- Create customer segments:
    - Champions
    - Loyal
    - At Risk

- Regular

- Identify churn-risk customers
- Export processed file for Power BI

```
rfm.head()
rfm.describe()
```

[16]

	last_purchase_date	frequency	monetary	recency
count	43233	43233.000000	43233.000000	43233.000000
mean	2023-04-11 20:01:40.922906112	2.313048	2105.364259	628.165499
min	2020-01-01 00:00:00	1.000000	5.150000	1.000000
25%	2022-05-21 00:00:00	1.000000	876.420000	225.000000
50%	2023-07-30 00:00:00	2.000000	1784.630000	519.000000
75%	2024-05-19 00:00:00	3.000000	2957.160000	954.000000
max	2024-12-29 00:00:00	10.000000	14558.680000	1825.000000
std	NaN	1.257115	1577.239778	478.122067

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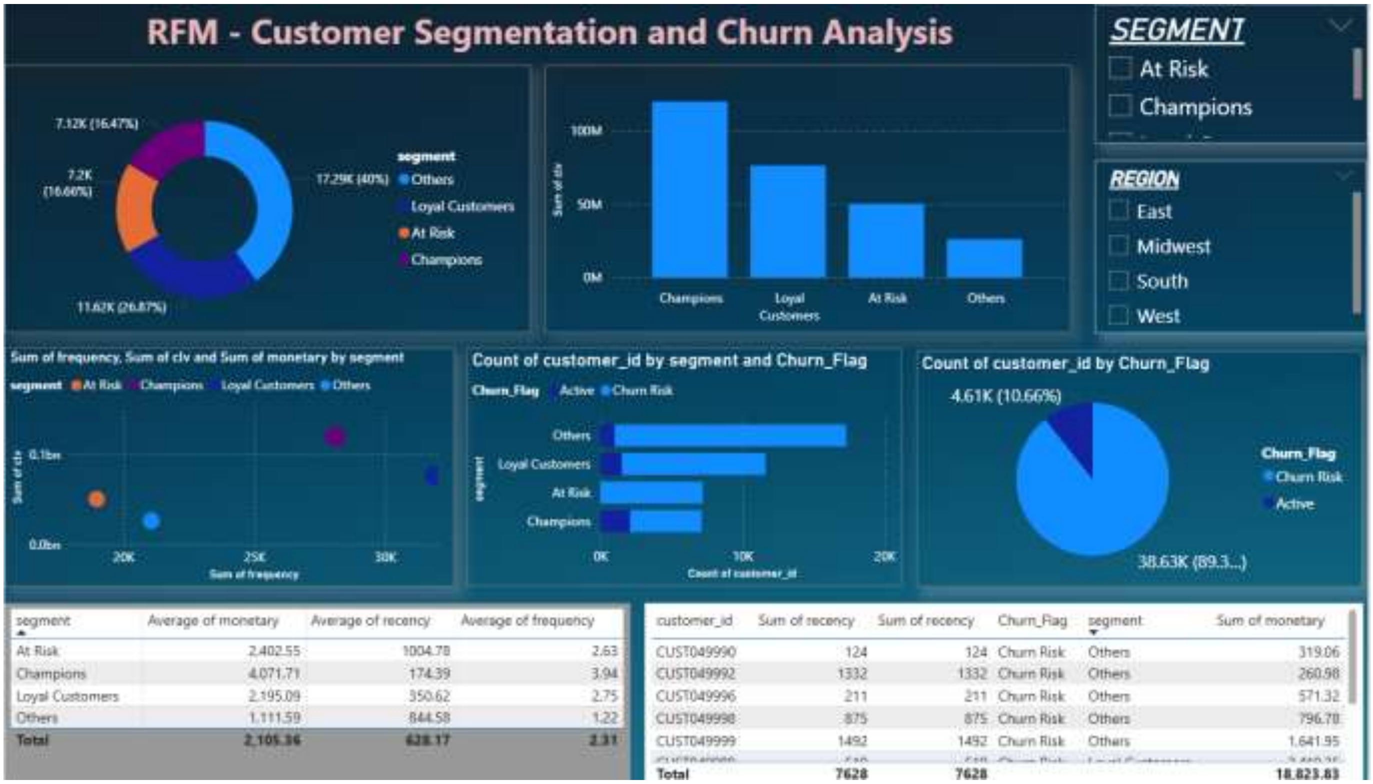
6. Power BI Dashboard

Interactive dashboards were created with:

- KPI cards (Orders, Customers, Sales)
- Sales by Category & Region
- Customer Segmentation charts
- RFM analysis visuals
- Churn risk distribution
- Drill-down (Country → Region)
- Slicers & filters
- DAX measures
- Row Level Security (RLS)



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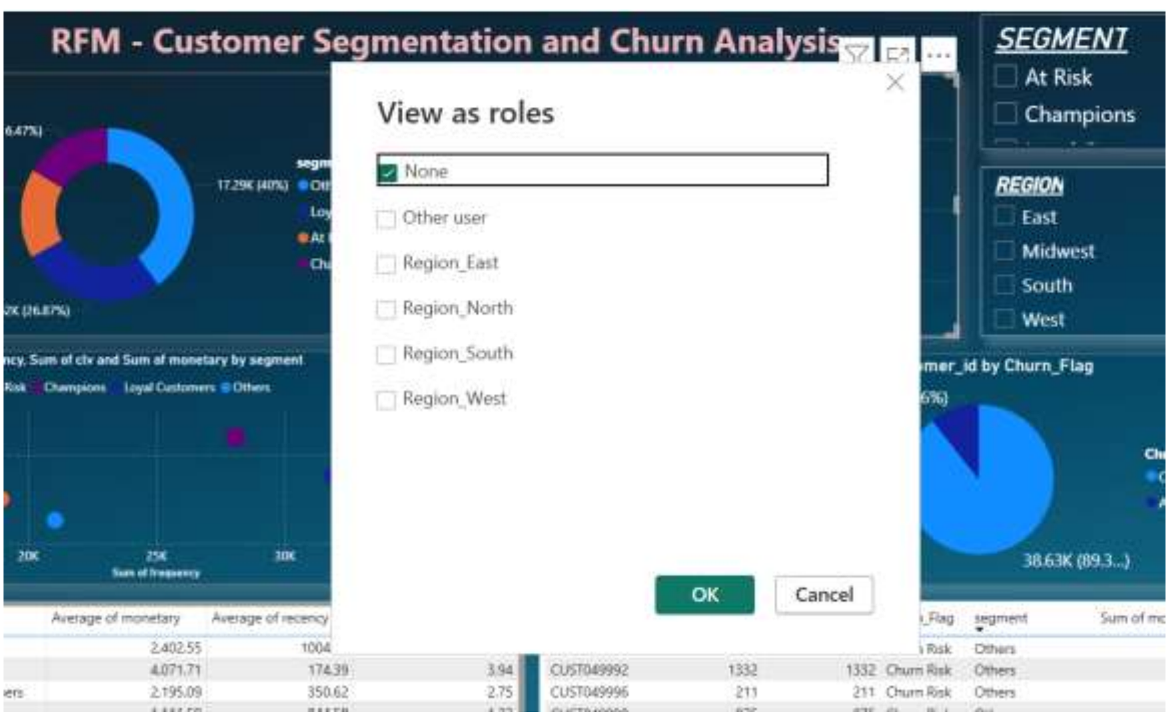
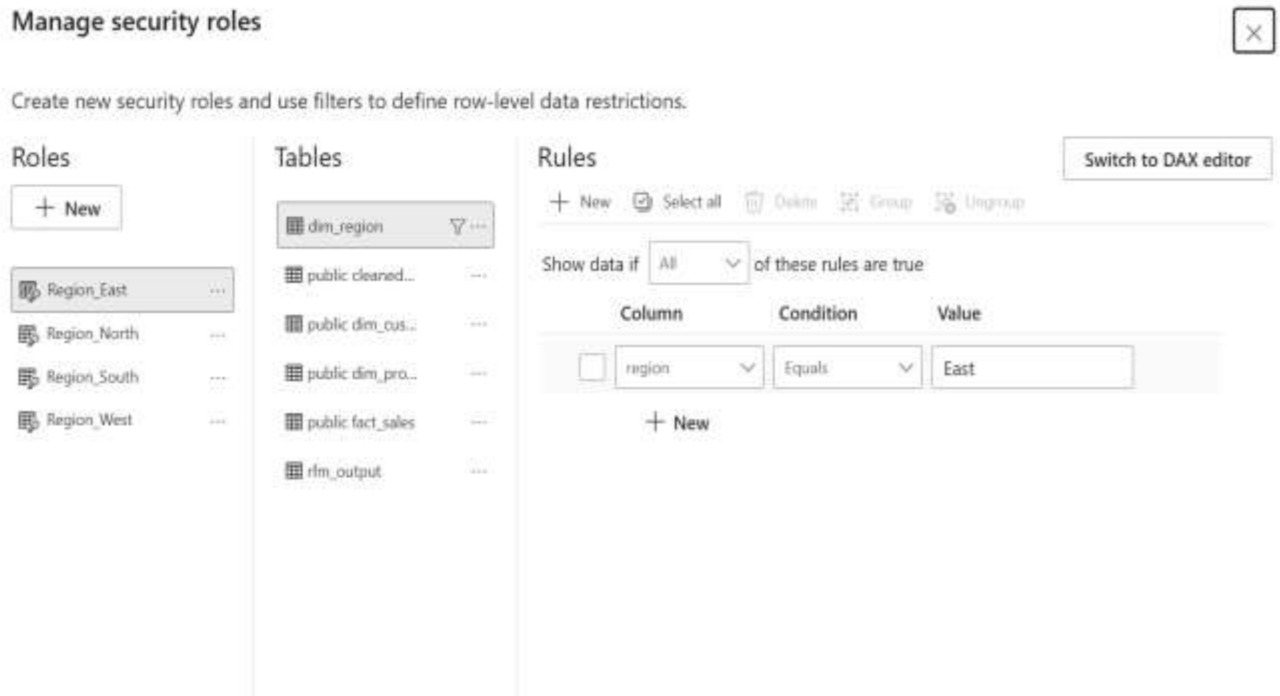
7. Row Level Security (RLS)

RLS was implemented to restrict dashboard access based on region.
Different users can view only their assigned country data.





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8. Automation Pipeline

Automation was implemented using:

- Python ETL script (rfm_pipeline.py)
- Automated RFM recalculation
- Output file refresh
- Windows Task Scheduler job
- Log file generation for execution tracking

This ensures updated analytics without manual effort.

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9. Key Insights

- High revenue comes from limited customer segments (Champions & Loyal)
- At-Risk customers identified using RFM recency patterns
- Category sales concentration visible
- Region-wise performance differences
- Customer retention opportunities identified

10. Business Impact

- Helps target high-value customers
- Supports churn prevention strategies
- Enables data-driven marketing
- Reduces manual reporting via automation
- Improves decision-making visibility

- Github link: https://github.com/Gaurii15/Consumer360-Retail-Analytics-Customer-Segmentation-Churn-Analysis